

TVMaze Shows API Web Scraping

Extracting TV Show Information using Python Requests API

Technology Used: Python | Requests | JSON | Pandas



Import Required Libraries

Import the Python libraries needed to send HTTP requests and organize the retrieved data.

```
In [28]: import requests  
import pandas as pd
```



Send HTTP GET Request

Use the Requests library to send a GET request and retrieve data from the API.

```
In [29]: url = "https://api.tvmaze.com/shows"  
response = requests.get(url)
```

```
In [30]: response.raise_for_status()
```

Parse JSON Response

Convert the JSON response received from the API into Python objects for further processing.

```
In [31]: shows = response.json()
```

```
In [32]: data = []
```

Explore Retrieved Data

Inspect the extracted JSON response to understand the available fields, nested structures, and shows-related information.

```
In [33]: for show in shows[:50]:      # First 50 shows
         rating = show.get("rating", {})

         data.append({
             "id": show.get("id", "N/A"),
             "name": show.get("name", "N/A"),
             "language": show.get("language", "N/A"),
             "genres": ", ".join(show.get("genres", [])),
             "rating": rating.get("average", "N/A") if rating else "N/A",
             "premiered": show.get("premiered", "N/A"),
             "image": show.get("image", {}).get("medium", "")
         })
```



Display Raw API Data

Print the JSON response to inspect the complete show information returned by the API.

```
In [34]: print(data)
```

```
[{'id': 1, 'name': 'Under the Dome', 'language': 'English', 'genres': 'Drama, Sci
ence-Fiction, Thriller', 'rating': 6.6, 'premiered': '2013-06-24', 'image': 'http
s://static.tvmaze.com/uploads/images/medium_portrait/610/1525272.jpg'}, {'id': 2,
'name': 'Person of Interest', 'language': 'English', 'genres': 'Action, Crime, Sc
ience-Fiction', 'rating': 8.8, 'premiered': '2011-09-22', 'image': 'https://stati
c.tvmaze.com/uploads/images/medium_portrait/163/407679.jpg'}, {'id': 3, 'name':
'Bitten', 'language': 'English', 'genres': 'Drama, Horror, Romance', 'rating': 7.
4, 'premiered': '2014-01-11', 'image': 'https://static.tvmaze.com/uploads/images/
medium_portrait/0/15.jpg'}, {'id': 4, 'name': 'Arrow', 'language': 'English', 'ge
nres': 'Drama, Action, Science-Fiction', 'rating': 7.4, 'premiered': '2012-10-1
0', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/143/35896
7.jpg'}, {'id': 5, 'name': 'True Detective', 'language': 'English', 'genres': 'Dr
ama, Crime, Thriller', 'rating': 8.1, 'premiered': '2014-01-12', 'image': 'http
s://static.tvmaze.com/uploads/images/medium_portrait/490/1226764.jpg'}, {'id': 6,
'name': 'The 100', 'language': 'English', 'genres': 'Action, Adventure, Science-F
iction', 'rating': 7.7, 'premiered': '2014-03-19', 'image': 'https://static.tvmaz
e.com/uploads/images/medium_portrait/477/1194981.jpg'}, {'id': 7, 'name': 'Homela
nd', 'language': 'English', 'genres': 'Drama, Thriller, Espionage', 'rating': 8.
2, 'premiered': '2011-10-02', 'image': 'https://static.tvmaze.com/uploads/images/
medium_portrait/498/1245275.jpg'}, {'id': 8, 'name': 'Glee', 'language': 'Englis
h', 'genres': 'Drama, Music, Romance', 'rating': 6.7, 'premiered': '2009-05-19',
'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/73.jpg'},
{'id': 9, 'name': 'Revenge', 'language': 'English', 'genres': 'Drama, Thriller, M
ystery', 'rating': 7.8, 'premiered': '2011-09-21', 'image': 'https://static.tvmaz
e.com/uploads/images/medium_portrait/82/206879.jpg'}, {'id': 10, 'name': 'Grimm',
'language': 'English', 'genres': 'Drama, Crime, Supernatural', 'rating': 8.4, 'pr
emiered': '2011-10-28', 'image': 'https://static.tvmaze.com/uploads/images/medium
_portrait/69/174906.jpg'}, {'id': 11, 'name': 'Gotham', 'language': 'English', 'g
enres': 'Drama, Action, Crime', 'rating': 7.7, 'premiered': '2014-09-22', 'imag
e': 'https://static.tvmaze.com/uploads/images/medium_portrait/189/474715.jpg'},
{'id': 12, 'name': 'Lost Girl', 'language': 'English', 'genres': 'Drama, Fantasy,
Horror', 'rating': 7.9, 'premiered': '2010-09-12', 'image': 'https://static.tvmaz
e.com/uploads/images/medium_portrait/0/137.jpg'}, {'id': 13, 'name': 'The Flash',
'language': 'English', 'genres': 'Drama, Action, Science-Fiction', 'rating': 7.5,
'premiered': '2014-10-07', 'image': 'https://static.tvmaze.com/uploads/images/med
ium_portrait/448/1121792.jpg'}, {'id': 14, 'name': 'Continuum', 'language': 'Engl
ish', 'genres': 'Drama, Crime, Science-Fiction', 'rating': 8, 'premiered': '2012-
05-27', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/184.
jpg'}, {'id': 15, 'name': 'Constantine', 'language': 'English', 'genres': 'Drama,
Action, Horror', 'rating': 7.5, 'premiered': '2014-10-24', 'image': 'https://stat
ic.tvmaze.com/uploads/images/medium_portrait/0/154.jpg'}, {'id': 16, 'name': 'Pen
ny Dreadful', 'language': 'English', 'genres': 'Horror, Thriller, Supernatural',
'rating': 8.2, 'premiered': '2014-05-11', 'image': 'https://static.tvmaze.com/upl
oads/images/medium_portrait/48/122260.jpg'}, {'id': 18, 'name': 'The Amazing Rac
e', 'language': 'English', 'genres': 'Action, Adventure, Family', 'rating': 6.8,
'premiered': '2001-09-05', 'image': 'https://static.tvmaze.com/uploads/images/med
ium_portrait/587/1468637.jpg'}, {'id': 19, 'name': 'Supernatural', 'language': 'E
nglish', 'genres': 'Drama, Action, Supernatural', 'rating': 8.3, 'premiered': '20
05-09-13', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/44
5/1114097.jpg'}, {'id': 20, 'name': 'The Strain', 'language': 'English', 'genre
s': 'Drama, Horror, Thriller', 'rating': 7.5, 'premiered': '2014-07-13', 'image':
'https://static.tvmaze.com/uploads/images/medium_portrait/444/1111710.jpg'}, {'i
d': 21, 'name': 'The Last Ship', 'language': 'English', 'genres': 'Drama, Action,
Thriller', 'rating': 7.5, 'premiered': '2014-06-22', 'image': 'https://static.tvm
aze.com/uploads/images/medium_portrait/164/412464.jpg'}, {'id': 22, 'name': 'True
Blood', 'language': 'English', 'genres': 'Drama, Romance, Supernatural', 'ratin
g': 7.9, 'premiered': '2008-09-07', 'image': 'https://static.tvmaze.com/uploads/i
mages/medium_portrait/410/1026956.jpg'}, {'id': 23, 'name': 'Once Upon a Time in
Wonderland', 'language': 'English', 'genres': 'Drama, Adventure, Fantasy', 'ratin
g': 6.9, 'premiered': '2013-10-10', 'image': 'https://static.tvmaze.com/uploads/i
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mages/medium_portrait/73/183661.jpg'}, {'id': 24, 'name': 'Hawaii Five-0', 'language': 'English', 'genres': 'Drama, Action, Crime', 'rating': 7.7, 'premiered': '2010-09-20', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/213/532575.jpg'}, {'id': 25, 'name': 'Hellsing', 'language': 'Japanese', 'genres': 'Anime, Horror, Supernatural', 'rating': 7.3, 'premiered': '2001-10-10', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/178/446544.jpg'}, {'id': 26, 'name': 'Hellsing Ultimate', 'language': 'Japanese', 'genres': 'Drama, Action, Anime, Horror', 'rating': 7.3, 'premiered': '2006-02-10', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/178/446780.jpg'}, {'id': 27, 'name': 'Berserk', 'language': 'Japanese', 'genres': 'Anime, Fantasy, Horror', 'rating': 8, 'premiered': '1997-10-07', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/396/991619.jpg'}, {'id': 28, 'name': 'Californication', 'language': 'English', 'genres': 'Drama, Comedy', 'rating': 7.8, 'premiered': '2007-08-13', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/38/95017.jpg'}, {'id': 29, 'name': 'Vikings', 'language': 'English', 'genres': 'Drama, Action, History', 'rating': 8.6, 'premiered': '2013-03-03', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/286/715906.jpg'}, {'id': 30, 'name': 'American Horror Story', 'language': 'English', 'genres': 'Drama, Horror, Thriller', 'rating': 7.5, 'premiered': '2011-10-05', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/473/1183640.jpg'}, {'id': 31, 'name': 'Marvel's Agents of S.H.I.E.L.D.', 'language': 'English', 'genres': 'Action, Adventure, Science-Fiction', 'rating': 8, 'premiered': '2013-09-24', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/486/1215694.jpg'}, {'id': 32, 'name': ' Fargo', 'language': 'English', 'genres': 'Drama, Comedy, Crime', 'rating': 8.6, 'premiered': '2014-04-15', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/487/1219631.jpg'}, {'id': 33, 'name': 'Hemlock Grove', 'language': 'English', 'genres': 'Drama, Horror, Thriller', 'rating': 6.9, 'premiered': '2013-04-19', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/305.jpg'}, {'id': 34, 'name': 'Helix', 'language': 'English', 'genres': 'Horror, Science-Fiction, Thriller', 'rating': 6.5, 'premiered': '2014-01-10', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/126/316697.jpg'}, {'id': 35, 'name': 'The Killing', 'language': 'English', 'genres': 'Drama, Crime, Thriller', 'rating': 8.2, 'premiered': '2011-04-03', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/216/540157.jpg'}, {'id': 37, 'name': 'Intruders', 'language': 'English', 'genres': 'Drama, Fantasy, Thriller', 'rating': 7.1, 'premiered': '2014-08-23', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/126/316698.jpg'}, {'id': 38, 'name': 'Z Nation', 'language': 'English', 'genres': 'Action, Horror, Science-Fiction', 'rating': 7.2, 'premiered': '2014-09-12', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/170/426759.jpg'}, {'id': 39, 'name': 'Resurrection', 'language': 'English', 'genres': 'Drama, Science-Fiction, Mystery', 'rating': 7.1, 'premiered': '2014-03-09', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/350.jpg'}, {'id': 40, 'name': 'Death Note', 'language': 'Japanese', 'genres': 'Anime, Fantasy, Thriller', 'rating': 8.8, 'premiered': '2006-10-03', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/499/1249019.jpg'}, {'id': 41, 'name': 'Last Man Standing', 'language': 'English', 'genres': 'Comedy, Family', 'rating': 7.8, 'premiered': '2011-10-11', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/419/1049994.jpg'}, {'id': 42, 'name': 'Sleepy Hollow', 'language': 'English', 'genres': 'Drama, Mystery, Supernatural', 'rating': 7.5, 'premiered': '2013-09-16', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/81/204166.jpg'}, {'id': 43, 'name': 'Outlander', 'language': 'English', 'genres': 'Adventure, Romance, Science-Fiction', 'rating': 7.9, 'premiered': '2014-08-09', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/616/1541142.jpg'}, {'id': 44, 'name': 'Scorpion', 'language': 'English', 'genres': 'Drama, Action, Thriller', 'rating': 7.3, 'premiered': '2014-09-22', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/128/322484.jpg'}, {'id': 45, 'name': 'NCIS: New Orleans', 'language': 'English', 'genres': 'Drama, Crime', 'rating': 7.5, 'premiered': '2014-09-23', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/280/701985.jpg'}, {'id': 46, 'name': 'Forever', 'language': 'English', 'genres': 'Drama, Crime, Supernatural', 'rating': 8.1, 'premiered': '2014-09-23', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/280/701985.jpg'}

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014-09-22', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/383.jpg'}, {'id': 47, 'name': 'Witches of East End', 'language': 'English', 'genres': 'Drama, Fantasy, Romance', 'rating': 7.5, 'premiered': '2013-10-06', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/392.jpg'}, {'id': 48, 'name': 'Madam Secretary', 'language': 'English', 'genres': 'Drama', 'rating': 7.7, 'premiered': '2014-09-21', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/214/536175.jpg'}, {'id': 49, 'name': 'Brooklyn Nine-Nine', 'language': 'English', 'genres': 'Comedy, Action, Crime', 'rating': 8.1, 'premiered': '2013-09-17', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/402/1007484.jpg'}, {'id': 50, 'name': 'The Lottery', 'language': 'English', 'genres': 'Drama, Science-Fiction, Thriller', 'rating': 6.5, 'premiered': '2014-07-20', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/146/366444.jpg'}, {'id': 51, 'name': 'The Knick', 'language': 'English', 'genres': 'Drama, Medical', 'rating': 8.3, 'premiered': '2014-08-08', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/0/417.jpg'}, {'id': 52, 'name': 'How to Get Away with Murder', 'language': 'English', 'genres': 'Drama, Thriller, Legal', 'rating': 7.6, 'premiered': '2014-09-25', 'image': 'https://static.tvmaze.com/uploads/images/medium_portrait/212/531512.jpg'}]
```

Convert data into DataFrame

Transform the extracted shows data into a structured Pandas DataFrame for easier analysis and visualization.

```
In [35]: df = pd.DataFrame(data)
df
```

Out[35]:

	id	name	language	genres	rating	premiered	
0	1	Under the Dome	English	Drama, Science-Fiction, Thriller	6.6	2013-06-24	https://static.tvmaze.com
1	2	Person of Interest	English	Action, Crime, Science-Fiction	8.8	2011-09-22	https://static.tvmaze.com
2	3	Bitten	English	Drama, Horror, Romance	7.4	2014-01-11	https://static.tvmaze.com
3	4	Arrow	English	Drama, Action, Science-Fiction	7.4	2012-10-10	https://static.tvmaze.com
4	5	True Detective	English	Drama, Crime, Thriller	8.1	2014-01-12	https://static.tvmaze.com
5	6	The 100	English	Action, Adventure, Science-Fiction	7.7	2014-03-19	https://static.tvmaze.com
6	7	Homeland	English	Drama, Thriller, Espionage	8.2	2011-10-02	https://static.tvmaze.com
7	8	Glee	English	Drama, Music, Romance	6.7	2009-05-19	https://static.tvmaze.com
8	9	Revenge	English	Drama, Thriller, Mystery	7.8	2011-09-21	https://static.tvmaze.com
9	10	Grimm	English	Drama, Crime, Supernatural	8.4	2011-10-28	https://static.tvmaze.com
10	11	Gotham	English	Drama, Action, Crime	7.7	2014-09-22	https://static.tvmaze.com
11	12	Lost Girl	English	Drama, Fantasy, Horror	7.9	2010-09-12	https://static.tvmaze.com
12	13	The Flash	English	Drama, Action, Science-Fiction	7.5	2014-10-07	https://static.tvmaze.com

	id	name	language	genres	rating	premiered	
13	14	Continuum	English	Drama, Crime, Science-Fiction	8.0	2012-05-27	https://static.tvmaze.com
14	15	Constantine	English	Drama, Action, Horror	7.5	2014-10-24	https://static.tvmaze.com
15	16	Penny Dreadful	English	Horror, Thriller, Supernatural	8.2	2014-05-11	https://static.tvmaze.com
16	18	The Amazing Race	English	Action, Adventure, Family	6.8	2001-09-05	https://static.tvmaze.com
17	19	Supernatural	English	Drama, Action, Supernatural	8.3	2005-09-13	https://static.tvmaze.com
18	20	The Strain	English	Drama, Horror, Thriller	7.5	2014-07-13	https://static.tvmaze.com
19	21	The Last Ship	English	Drama, Action, Thriller	7.5	2014-06-22	https://static.tvmaze.com
20	22	True Blood	English	Drama, Romance, Supernatural	7.9	2008-09-07	https://static.tvmaze.com
21	23	Once Upon a Time in Wonderland	English	Drama, Adventure, Fantasy	6.9	2013-10-10	https://static.tvmaze.com
22	24	Hawaii Five-0	English	Drama, Action, Crime	7.7	2010-09-20	https://static.tvmaze.com
23	25	Hellsing	Japanese	Anime, Horror, Supernatural	7.3	2001-10-10	https://static.tvmaze.com
24	26	Hellsing Ultimate	Japanese	Drama, Action, Anime, Horror	7.3	2006-02-10	https://static.tvmaze.com
25	27	Berserk	Japanese	Anime, Fantasy, Horror	8.0	1997-10-07	https://static.tvmaze.com
26	28	Californication	English	Drama, Comedy	7.8	2007-08-13	https://static.tvmaze.com
27	29	Vikings	English	Drama, Action, History	8.6	2013-03-03	https://static.tvmaze.com

	id	name	language	genres	rating	premiered	
28	30	American Horror Story	English	Drama, Horror, Thriller	7.5	2011-10-05	https://static.tvmaze.com
29	31	Marvel's Agents of S.H.I.E.L.D.	English	Action, Adventure, Science-Fiction	8.0	2013-09-24	https://static.tvmaze.com
30	32	Fargo	English	Drama, Comedy, Crime	8.6	2014-04-15	https://static.tvmaze.com
31	33	Hemlock Grove	English	Drama, Horror, Thriller	6.9	2013-04-19	https://static.tvmaze.com
32	34	Helix	English	Horror, Science-Fiction, Thriller	6.5	2014-01-10	https://static.tvmaze.com
33	35	The Killing	English	Drama, Crime, Thriller	8.2	2011-04-03	https://static.tvmaze.com
34	37	Intruders	English	Drama, Fantasy, Thriller	7.1	2014-08-23	https://static.tvmaze.com
35	38	Z Nation	English	Action, Horror, Science-Fiction	7.2	2014-09-12	https://static.tvmaze.com
36	39	Resurrection	English	Drama, Science-Fiction, Mystery	7.1	2014-03-09	https://static.tvmaze.com
37	40	Death Note	Japanese	Anime, Fantasy, Thriller	8.8	2006-10-03	https://static.tvmaze.com
38	41	Last Man Standing	English	Comedy, Family	7.8	2011-10-11	https://static.tvmaze.com
39	42	Sleepy Hollow	English	Drama, Mystery, Supernatural	7.5	2013-09-16	https://static.tvmaze.com
40	43	Outlander	English	Adventure, Romance, Science-Fiction	7.9	2014-08-09	https://static.tvmaze.com
41	44	Scorpion	English	Drama, Action, Thriller	7.3	2014-09-22	https://static.tvmaze.com

	id	name	language	genres	rating	premiered	
42	45	NCIS: New Orleans	English	Drama, Crime	7.5	2014-09-23	https://static.tvmaze.com
43	46	Forever	English	Drama, Crime, Supernatural	8.1	2014-09-22	https://static.tvmaze.com
44	47	Witches of East End	English	Drama, Fantasy, Romance	7.5	2013-10-06	https://static.tvmaze.com
45	48	Madam Secretary	English	Drama	7.7	2014-09-21	https://static.tvmaze.com
46	49	Brooklyn Nine-Nine	English	Comedy, Action, Crime	8.1	2013-09-17	https://static.tvmaze.com
47	50	The Lottery	English	Drama, Science-Fiction, Thriller	6.5	2014-07-20	https://static.tvmaze.com
48	51	The Knick	English	Drama, Medical	8.3	2014-08-08	https://static.tvmaze.com
49	52	How to Get Away with Murder	English	Drama, Thriller, Legal	7.6	2014-09-25	https://static.tvmaze.com

In []:

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In []: